Chapter 10

10. Analysis of Variance.

1. Introduction and model.
2. Estimation.
3. Parametrization.
4. Tests.
5. Checking model assumptions.
6. Variance stabilizing transformation.
7. Further testing problems.

Introduction and model

A common statistical problem is the comparison of 2 or more populations. The overall objective is to analyse whether the populations are different in certain characteristics.

Example. We would like to compare k different medical treatments. The i-th treatment is applied to n_i individuals $\Longrightarrow$ the total sample size is $n = n_1 + \cdots + n_k$.

Terminology: We have some factor (treatment) with k levels (treatment 1, treatment 2, ... , treatment k).

Our data set consists of k groups or cells where each group contains observations which correspond to a certain level of the factor.

For the moment we consider only one factor $\Longrightarrow$ 1-way ANOVA.

Introduction and model

We have n_i observations at level i $(1 \leq i \leq k) \Longrightarrow$

$$Y_{i1}, \ldots, Y_{in_i} \quad \text{group } i.$$

General assumption:

$$Y_{ij} \sim N(\beta_i, \sigma^2)$$
$$Y_{ij} \quad \text{are independent for all } i, j.$$

Our principal goal is to test $H_0 : \beta_1 = \ldots = \beta_k$ against $H_1 : \beta_\ell \neq \beta_m$ for at least one pair (m, ℓ).

Introduction and model

We can treat the problem in the context of the general linear model:

$$\begin{pmatrix} Y_{11} \\ \vdots \\ Y_{1n_1} \\ Y_{21} \\ \vdots \\ Y_{2n_2} \\ \vdots \\ Y_{k1} \\ \vdots \\ Y_{kn_k} \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_2 \\ \vdots \\ \beta_k \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \varepsilon_{11} \\ \vdots \\ \varepsilon_{1n_1} \\ \varepsilon_{21} \\ \vdots \\ \varepsilon_{2n_2} \\ \vdots \\ \varepsilon_{k1} \\ \vdots \\ \varepsilon_{kn_k} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} \times \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \varepsilon_{11} \\ \vdots \\ \varepsilon_{1n_1} \\ \varepsilon_{21} \\ \vdots \\ \varepsilon_{2n_2} \\ \vdots \\ \varepsilon_{k1} \\ \vdots \\ \varepsilon_{kn_k} \end{pmatrix}.$$

Introduction and model

We remark that our design matrix

$$X = \begin{pmatrix} 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}, \tag{1}$$

has orthogonal columns $e_1, \dots, e_k$ with $\|e_i\|^2 = n_i$.

Estimation

Applying our general results we obtain

$$\hat{\beta} = (X'X)^{-1}X'Y = \cdots = (\bar{Y}_1, \ldots, \bar{Y}_k)',$$

where $\bar{Y}_i$ is the mean of a cell.

We have

$$\hat{\beta} \sim \mathcal{N}_k(\beta, \sigma^2(X'X)^{-1}).$$

Here we get

$$(X'X)^{-1} = \mathrm{diag}(\frac{1}{n_1}, \ldots, \frac{1}{n_k}),$$

and hence $\hat{\beta}_i \overset{\text{indep}}{\sim} N(\beta_i, \frac{\sigma^2}{n_i})$.

Estimation

Applying our general results we obtain furthermore that

$$\tilde{\sigma}^2 = \frac{1}{n-k} SSR = \frac{1}{n-k} \|Y - \hat{\mu}\|^2 \sim \frac{\sigma^2}{n-k} \chi^2_{n-k}.$$

And hence

$$\tilde{\sigma}^2 = MSR = \frac{1}{n-k} \sum_{i=1}^{k} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2.$$

Parametrization

Note that often the model is stated in a reparametrized form. This means that the basis represented by the design is changed.

E.g. we could use the design matrix which puts the intercept explicitly in the first column:

$$X_0 = \begin{pmatrix} 1 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \dots & 0 \\ 1 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \dots & 1 \end{pmatrix} = (1_n, e_2, \dots, e_k).$$

Parametrization

In this case, note that

$$X = X_0 G \quad \text{with} \quad G = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ -1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ldots & \vdots \\ -1 & 0 & \cdots & 1 \end{pmatrix}.$$

Hence $X\beta = X_0 G\beta =: X_0\gamma$ with

$$\gamma = (\beta_1, \beta_2 - \beta_1, \ldots, \beta_k - \beta_1)'.$$

So in this parametrization the first coefficient is the mean of the first group (control group) and the other coefficients relate to the deviations from control group.

Parametrization

More generally we use $X_C = XM_C$ where $M_C = [1_k, C] \in \mathbb{R}^{k \times k}$.
Here C is a $k \times (k-1)$ matrix called contrast matrix.

By default we have in R

$$C = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \in \mathbb{R}^{k \times (k-1)}.$$

yielding the design matrix X_0.

We have the following parametrization:

$$X\beta = XM_C M_C^{-1}\beta = X_C\beta_C = XM_C\beta_C.$$

Parametrization

We can change the contrast matrix in R. Let us check the default contrast first:

```
> fac <- factor(rep(c("A","B","C"),c(2,3,4)))
> contrasts(fac)
  B C
A 0 0
B 1 0
C 0 1
```

We switch now to a different contrast:

```
> contrasts(fac) <- "contr.sum"
> contrasts(fac)
  [,1] [,2]
A    1    0
B    0    1
C   -1   -1
```

Parametrization

Let us denote the corresponding parameters with

$$\beta_C = (\mu, \alpha_1, \ldots, \alpha_{k-1})'.$$

Then due to

$$M_C \beta_C = \beta$$

we easily see that

$$\beta = (\mu + \alpha_1, \ldots, \mu + \alpha_{k-1}, \mu - (\alpha_1 + \cdots + \alpha_{k-1}))'.$$

So in this parametrization the first coefficient μ is some base level, and the coefficients α_i are additive effects related to the i-th group.

Parametrization

Letting $m = \frac{1}{k}(\bar{Y}_1 + \cdots + \bar{Y}_k)$ (mean of the means) one can show (exercise) that

$$\hat{\beta}_C = (m, \bar{Y}_1 - m, \ldots, \bar{Y}_{k-1} - m)'.$$

Note: The mean of the means m is NOT the grand mean $\bar{Y}$, unless we have a balanced design.

Tests

We first consider the test

$$H_0 : \beta_1 = \ldots = \beta_k \quad \text{against} \quad H_1 : \beta_m \neq \beta_\ell \quad \text{for some pair } (m, \ell).$$

Let $L_2 = \text{span}\{1_n\}$ where $1_n = (1, \ldots, 1)'$ and $L_1 = \text{span}\{e_1, \ldots, e_k\}$. Then

$$\hat{\mu}_1 = (\underbrace{\bar{Y}_1, \ldots, \bar{Y}_1}_{n_1}, \ldots, \underbrace{\bar{Y}_k, \ldots, \bar{Y}_k}_{n_k})'$$

and

$$\hat{\mu}_2 = (\underbrace{\bar{Y}, \ldots, \bar{Y}}_{n})'.$$

Tests

Hence,

$$SSA := \|\hat{\mu}_1 - \hat{\mu}_2\|^2 = \sum_{i=1}^{k} n_i(\bar{Y}_i - \bar{Y})^2.$$

SSA is called sums of squares between groups (treatments).

Tests

Consequently,

$$F(Y) = \frac{\frac{1}{k-1}\sum_{i=1}^{k} n_i(\bar{Y}_i - \bar{Y})^2}{\frac{1}{n-k}\sum_{i=1}^{k}\sum_{j=1}^{n_i}(Y_{ij} - \bar{Y}_i)^2}$$

$$= \frac{\frac{1}{k-1}SSA}{\frac{1}{n-k}SSR} = \frac{MSA}{MSR} \sim F_{k-1,n-k}.$$

We reject H_0 if $F(Y) > F_{k-1,n-k;1-\alpha}$.

Example

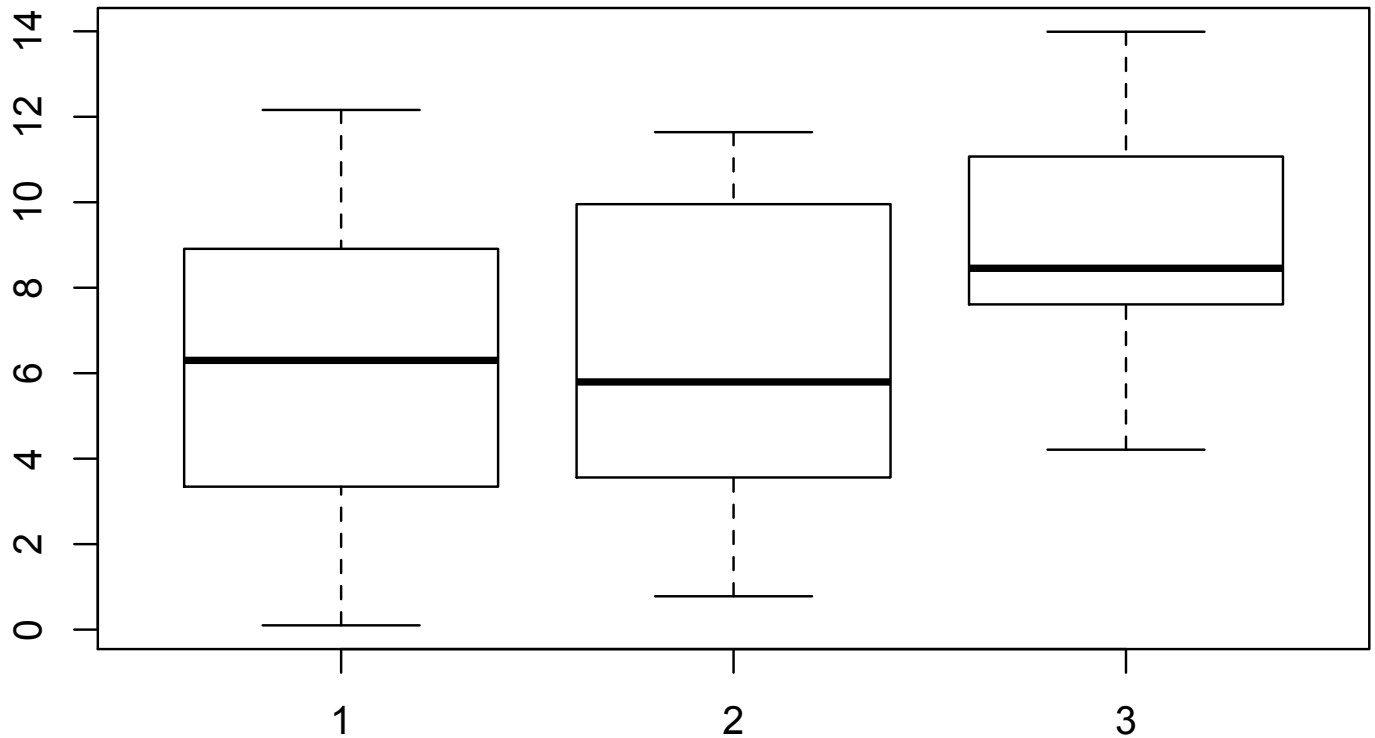
The file `battery.dat` contains 36 data points. Each data point is 3 dimensional and has in its first component `Watt-hours` of a battery, in the second component a type of an `additive` (1 = CONTROL, 2 = AD-X2, 3 = MG-NA) in the third component the battery ID (19–30).

```
     Watt Additive ID
1    6.12        1 19
2    9.13        1 20
3   12.16        1 21
4    0.92        1 22
5   10.07        1 23
....
```

We import the data to `R` and call the data.frame `battery`. We use the function `as.factor()` in ordre to transform the variable `Additive` to factor type.

Example

```
> boxplot(Watt~Additive, data=battery)
```



Example

The boxplots show no outliers and are reasonably balanced so that there is no obvious violation of the normality and homogeneous variance assumption.

We perform a 1-way ANOVA using the function aov in R.

```
> battery.aov <- aov(Watt~Additive, data=battery)
> summary(battery.aov)
            Df Sum Sq Mean Sq F value Pr(>F)
Additive     2   67.0   33.51   2.982 0.0645 .
Residuals   33  370.9   11.24
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> battery.aov$coefficients
(Intercept)   Additive2   Additive3
   6.090000    0.370000    3.061667
```

Example

The ANOVA table contains the following information

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Additive	k-1	SSA	MSA	MSA/MSR	P(F>f)
Residuals	n-k	SSR	MSR		

As it has been mentioned before, R parametrizes by default in the form

$$(\beta_1, \beta_2 - \beta_1, \ldots, \beta_k - \beta_1)$$

In our setting the coefficients are given in the form

$$(\hat{\beta}_1, \hat{\beta}_2 - \hat{\beta}_1, \hat{\beta}_3 - \hat{\beta}_1) = (\bar{Y}_1, \bar{Y}_2 - \bar{Y}_1, \bar{Y}_3 - \bar{Y}_1).$$

Example

Alternatively, we could also use the function `lm` for the ANOVA:

```
> battery.lm=lm(Watt~Additive, data=battery)
> summary(battery.lm)

Residuals:
    Min      1Q  Median      3Q     Max
-5.9900 -2.0762 -0.5508  2.6475  6.0700

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.0900     0.9678   6.293 4.09e-07 ***
Additive2     0.3700     1.3687   0.270   0.7886
Additive3     3.0617     1.3687   2.237   0.0322 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.353 on 33 degrees of freedom
Multiple R-squared:  0.153,Adjusted R-squared:  0.1017
F-statistic: 2.982 on 2 and 33 DF,  p-value: 0.06452
```

Example

To get the original parametrization one might be tempted to remove the intercept, but this is not a good idea ...

```
> battery.lm <- lm(Watt~Additive-1, data=bat)
> summary(battery.lm)

Coefficients:
          Estimate Std. Error t value Pr(>|t|)
Additive1    6.090      0.968    6.29  4.1e-07 ***
Additive2    6.460      0.968    6.67  1.3e-07 ***
Additive3    9.152      0.968    9.46  6.4e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.35 on 33 degrees of freedom
Multiple R-squared:  0.84,Adjusted R-squared:  0.826
F-statistic: 57.9 on 3 and 33 DF,  p-value: 3.11e-13
```

In this case the F-Test is not testing $L_1 = span(X)$ vs $L_2 = span(1_n)$, but $L_1 = span(X)$ vs $L_2 = span(0)$.

Example

In other words: removing the intercept means that we are not testing for (general) i.i.d. but for i.i.d. zero mean!

In this case

$$SST = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (Y_{ij} - 0)^2,$$

and

$$SSA = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (\bar{Y}_i - 0)^2.$$

This explains the big R^2: SST gets very big and will "kill" the (correct) SSR.

Important: even when the design matrix X already implicitly includes an intercept (like in (1)) `R` will still by default add the column 1_n to X. However, $\texttt{lm}(y \sim X - 1)$ doesn't solve the problem. It doesn't mean "don't add 1_n" it means, "do regression through 0".

Checking model assumptions

We can use a mix of diagnostic plots and statistical tests in order to verify the model assumptions.

Let us recall that the residual vector is $R = (I_n - H)\varepsilon$. Here $H = X(X'X)^{-1}X'$ has the simple form

$$H = \sum_{i=1}^{k} \frac{1}{n_i} e_i e_i'.$$

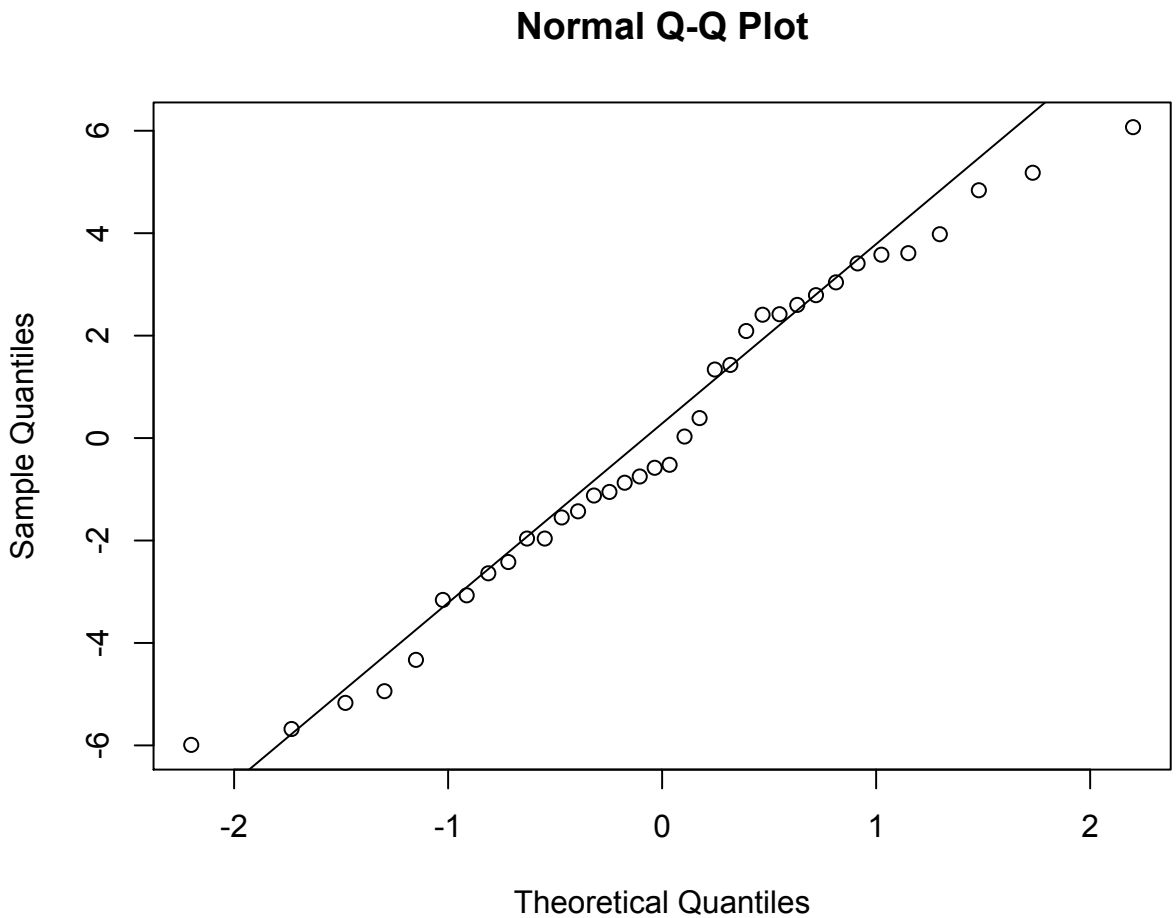
Hence $R \sim \mathcal{N}_n(0, \sigma^2(I_n - H))$ implies that $R_{ij} \sim N(0, \sigma^2(1 - \frac{1}{n_i}))$.

We have $\mathrm{Cov}(R_{ij}, R_{\ell m}) = -\frac{1}{n_i}$ if $\ell = i$ and zero else.

Thus, if the numbers n_i (number of observations in i-th group) are not too small, we expect the residuals to behave like i.i.d. $N(0, \sigma^2)$ variables and the standardized residuals $R_{ij}/\sqrt{MSR}$ should be behave like i.i.d. $N(0, 1)$ variables.

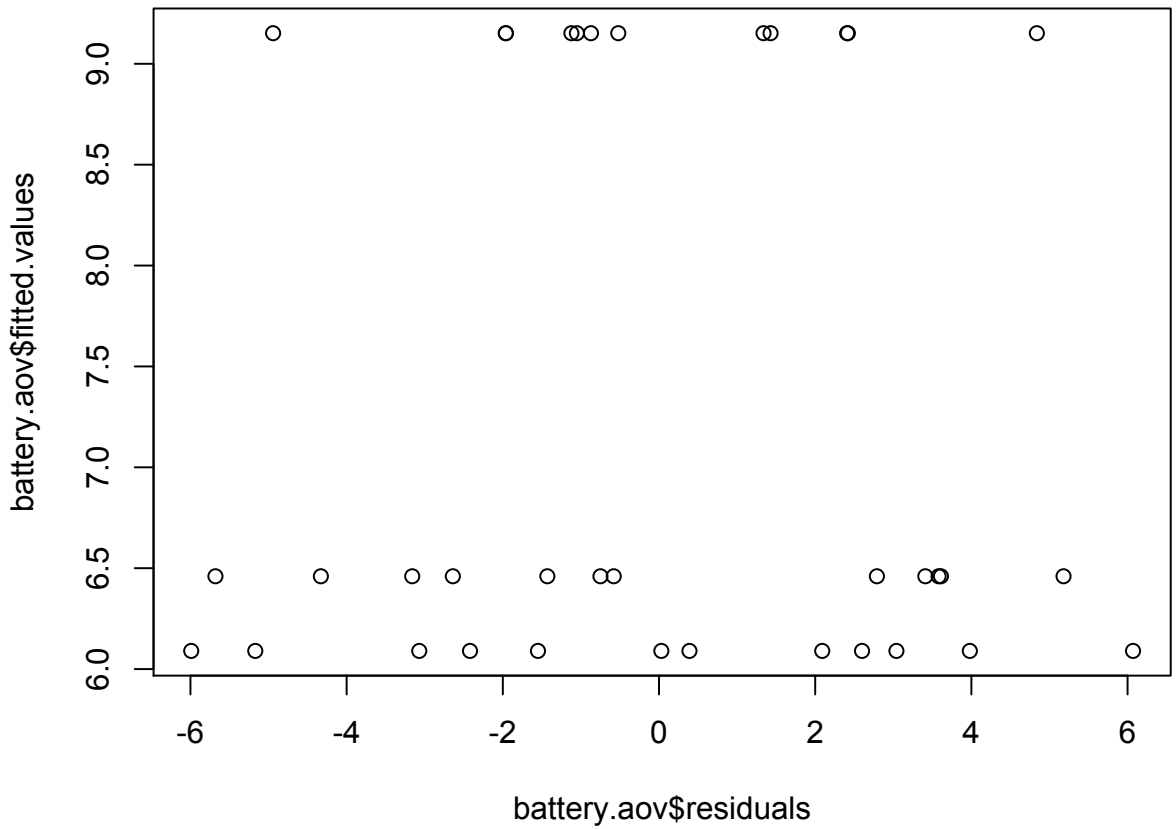
Checking model assumptions

To check normality assumption we consider a qq-plot:



Checking model assumptions

To check the assumpion of a homogenous variance (same variance in all groups) we plot residuals against fitted values:



Checking model assumptions

Alternatively we can run some tests instead of inspecting residuals visually.

A popular test for homogeneous variance is Levene's test. The idea is simple: we do standard 1-way ANOVA on the variables $Z_{ij} = |Y_{ij} - \bar{Y}_i|$. A theoretically better justified test is Bartlett's test (see Exercises) but its practical performance is less good, especially if the normality assumption is violated.

```
> summary(aov(abs(battery.aov$residuals)~battery$Additive))
                  Df Sum Sq Mean Sq F value Pr(>F)
battery$Additive   2   7.90   3.949   1.386  0.264
Residuals         33  94.03   2.849
```

The output of the test shows no significant violation of the null and we therefore consider the equal variance assumption as not being violated.

Variance stabilizing transformations

If the variances are not homogeneous in regression or ANOVA context we may think of transforming (apply some function to) the variable of interest.

In the following we will look at the class of Box-Cox transformations

$$T_\lambda(Y) = \begin{cases} \frac{Y^\lambda - 1}{\lambda}, & \text{for } \lambda \neq 0, \\ \log Y, & \text{for } \lambda = 0. \end{cases}$$

provided $Y > 0$. (Note that $\lim_{\lambda \to 0} T_\lambda(x) = T_0(x)$.)

Variance stabilizing transformations

More generally we look at

$$T_\lambda(Y) = \begin{cases} \frac{(Y+\lambda_2)^{\lambda_1}-1}{\lambda_1}, & \text{for } \lambda_1 \neq 0, \\ \log(Y+\lambda_2), & \text{for } \lambda_1 = 0. \end{cases}$$

provided $Y > -\lambda_2$.

Variance stabilizing transformations

The basic idea is this: assume that $\mathrm{Var}(Y) = \sigma^2(E(Y))^{2\alpha}$. Hence, the variance is proportional to some power of the mean.

By a Taylor expansion around $\mu = EY$ we get

$$T_\lambda(Y) \approx T_\lambda(\mu) + \frac{\partial}{\partial \mu} T_\lambda(\mu)(Y - \mu)$$

and hence

$$\mathrm{Var}(T_\lambda(Y)) \approx \left(\frac{\partial}{\partial \mu} T_\lambda(\mu)\right)^2 \times \mathrm{Var}(Y) = \sigma^2 \mu^{2(\lambda-1)+2\alpha},$$

which is constant if we choose $\lambda = 1 - \alpha$. If $\alpha = 1$, we get by the same argument, approximately constant variance by choosing T_0.

Variance stabilizing transformations

We can choose λ (note that α is unknown) by a likelihood approach (below). Nonetheless, it should be noted that often the transformation is chosen by visual inspection or such that we obtain an interpretable result.

Quite commonly $T(Y) = \sqrt{Y}$ or $T(Y) = \log(Y)$ are used.

Variance stabilizing transformations

How do we choose λ by MLE? The argument is a bit hand-waving:

Let $Y = (Y_1, \ldots, Y_n)'$ be our response vector and let $Z_\lambda = T_\lambda(Y) = (T_\lambda(Y_1), \ldots, T_\lambda(Y_n))'$. We assume that

$$Z_\lambda = X\beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}_n(0, \sigma^2 I_n). \tag{2}$$

Note that (2) cannot hold if $T_\lambda(x) = \sqrt{x}$ or $T_\lambda(x) = x^2$. When $\lambda > 0$ Bickel and Doksum (1981)[1] suggest extending the definition of the power family to

$$T_\lambda(x) = \frac{|x|^\lambda \text{sgn}(x) - 1}{\lambda}; \quad \lambda > 0, x \in \mathbb{R}. \tag{3}$$

Then, if in fact the Y's are positive, the likelihood function will be as outlined below (exercise).

[1] *An Analysis of Transformations Revisited,* JASA, Vol. 76, pp. 296–311.

Variance stabilizing transformations

Since we assume independent data, we can do the transformation component by component. Take component i:

$$P(T_\lambda(Y_i) \leq z) = \Phi((z - \mu_i)/\sigma).$$

Thus

$$F_{Y_i}(y) = P(Y_i \leq T_\lambda^{-1}(T_\lambda(y))) = \Phi((T_\lambda(y) - \mu_i)/\sigma)$$

and thus

$$f^{Y_i}(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(T_\lambda(y)-\mu_i)^2} \frac{\partial}{\partial y} T_\lambda(y).$$

Variance stabilizing transformations

With T_λ as in (3) we get

$$f^Y(y) = f^Y(y|\beta,\sigma,\lambda) = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n e^{-\frac{1}{2\sigma^2}\|T_\lambda(y)-X\beta\|^2} \prod_{i=1}^n |y_i|^{\lambda-1}.$$

For fixed λ, it can be easily seen that the MLE's for β and σ^2 are

$$\hat{\beta}(\lambda) = (X'X)^{-1}X'T_\lambda(Y)$$
$$\hat{\sigma}^2(\lambda) = \frac{1}{n}\|T_\lambda(Y) - X\hat{\beta}(\lambda)\|^2$$

and then the profile log-likelihood for λ is given as

$$L(\lambda) = \log f^Y(Y|\beta(\lambda),\sigma(\lambda),\lambda) = C - \frac{n}{2}\log\hat{\sigma}^2(\lambda) + (\lambda-1)\sum_{k=1}^n \log Y_i.$$

(Recall that we have been requiring $Y_i > 0$).

Variance stabilizing transformations

Example: The `R` library `MASS` contains a dataframe called `quine`. From the description file:

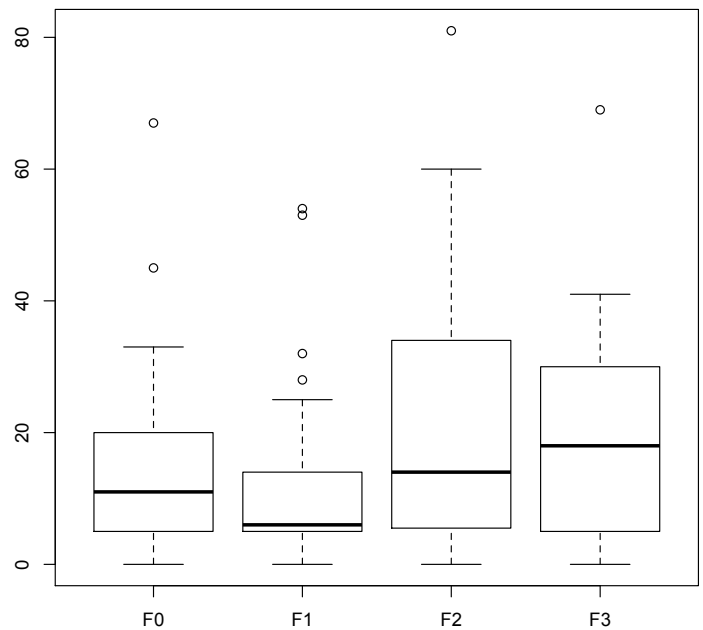
"The quine data frame has 146 rows and 5 columns. Children from Walgett, New South Wales, Australia, were classified by Culture, Age, Sex and Learner status and the number of days absent from school in a particular school year was recorded."

We would like to see if `Days` (absent) depends on `Age` (forms "F0", "F1," "F2" or "F3").

Variance stabilizing transformations

The boxplot `Days` vs. `Age` indicates that

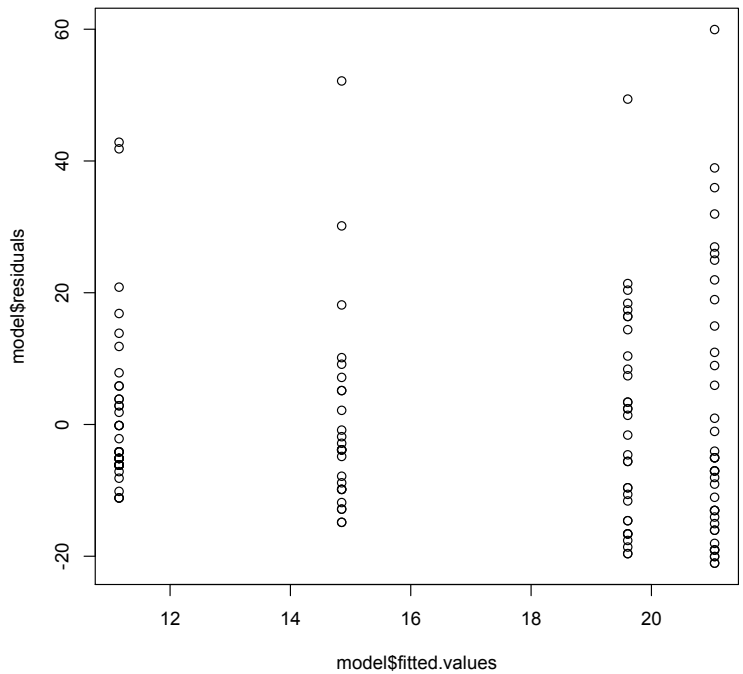
1. the homogeneous variance assumption may be violated,
2. data are right-skewed, so normality will be violated, too.



Variance stabilizing transformations

When we plot the group-means against residuals, we see that larger group means in tendency give larger variance:

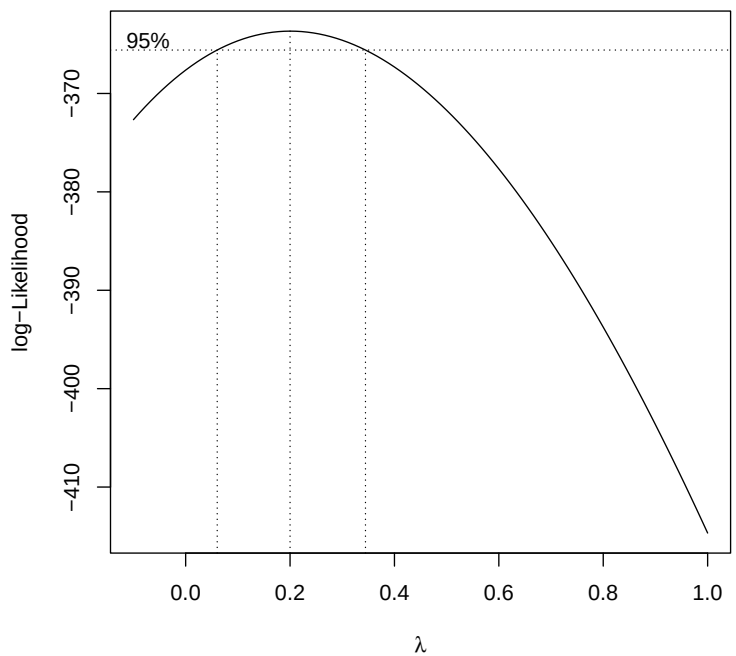
```
> model=aov(Days~Age,quine)
> plot(model$fitted.values, model$residuals)
```



Variance stabilizing transformations

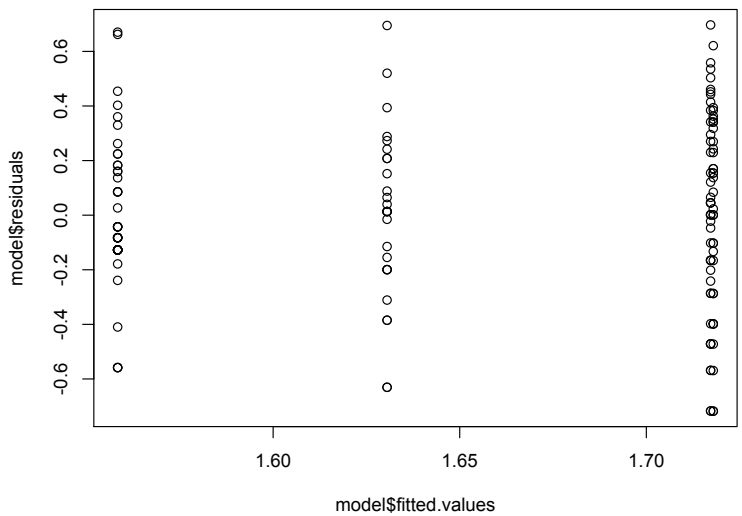
Since we have several zero values in each form we first do the transformation $Y \to Y + 1$. In this case we avoid troubles with the terms $\log Y_k$ which appear in estimation of λ.

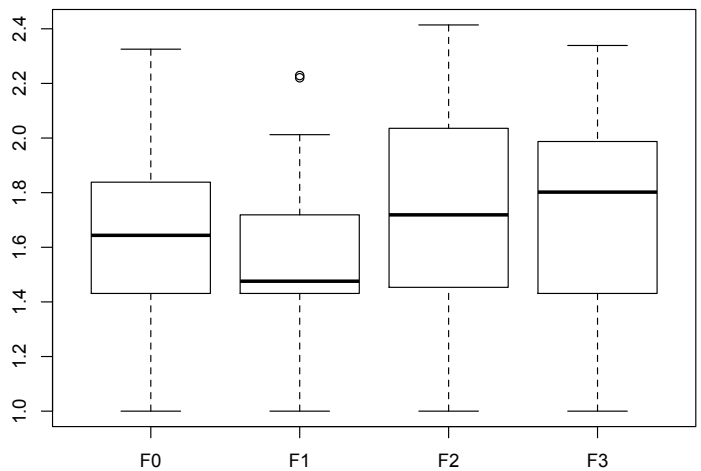
```
> y <- quine$Days+1
> x <- quine$Age
> boxcox(y~x, lambda=seq(-0.1,1,by=0.1))
```



Variance stabilizing transformations

```
> model <- aov((Days+1)^0.2~Age,quine)
> plot(model$fitted.values, model$residuals)
> boxplot((Days+1)^0.2~Age,quine)
```





It is interesting to see that the groups F3 and F4 move together.

This should also warn us, that we are actually doing inference on a different model!

Further testing problems

So far we have only considered the testing situation in which we test equality of all means. Other tests of interest are e.g. of the form

$$\beta_1 = \beta_3 \quad \text{or} \quad \beta_1 = \frac{\beta_2 + \beta_3}{2} \quad \text{or} \quad \frac{\beta_2 + \beta_3}{2} = \frac{\beta_1 + \beta_4 + \beta_5}{3}.$$

Remark

To maintain the correct size, such tests need to be planned before looking at the data! Otherwise we need to adjust the significance level. E.g. if we compare the smallest with the largest average. See discussion on post-hoc tests below.

Further testing problems

Example: Suppose we have two drugs for the same active substance. In a randomized study we divide $n = n_1 + n_2 + n_3$ patients in three groups and apply the placebo to first (control) group, drug A to the second group and drug B to the third group.

Question 1: Is the effect of drug A different from drug B?

$\Longleftrightarrow \mathcal{H}_0 : \beta_2 = \beta_3$?

Question 2: Does the active substance work?

$\Longleftrightarrow \mathcal{H}_0 : \beta_1 = \frac{\beta_2+\beta_3}{2}$.

Further testing problems

More generally, the hypotheses above can be stated as

$$c'\beta = 0 \quad \text{with} \quad 1_k' c = 0.$$

The vector c is called a *contrast.* Note that contrasts matrices defined previously don't require $1_k' C = 0$!

Lemma
Let us define $C(Y) = c'\hat{\beta}$. Under $H_0 : c'\beta = \mu_c$, we have

$$c'\hat{\beta} \sim N(\mu_c, \sigma^2 s_c^2) \quad \textit{where} \quad s_c^2 = \sum_{i=1}^{k} \frac{c_i^2}{n_i}.$$

Proof. Exercise. □

Further testing problems

Since $\tilde{\sigma}^2 \sim \sigma^2 \frac{1}{n-k}\chi^2_{n-k}$ is independent of $\hat{\beta}$ it follows under H_0 that

$$\frac{C(Y) - \mu_c}{\tilde{\sigma} s_c} \sim t_{n-k}.$$

We reject H_0 if

$$\mu_c \notin [C(Y) - \tilde{\sigma} s_c t_{n-k;1-\alpha}, C(Y) + \tilde{\sigma} s_c t_{n-k;1-\alpha}].$$

Further testing problems

Consider a balanced design, i.e. if $n_i = m$ for $i = 1, \ldots, k$. For orthogonal contrasts, i.e. $c_i \perp c_j, \quad i \neq j$ (we can choose up to $k-1$ such vectors) we obtain under

$$H_0 \colon c_i \beta = \mu_{c_i} \quad \text{for} \quad 1 \leq i \leq k-1$$

that the contrasts satisfy

$$S := (c_1' \hat{\beta}, \ldots, c_{k-1}' \hat{\beta})' \sim \mathcal{N}_{k-1}\big(\mu_c, \frac{\sigma^2}{m} \mathrm{diag}(\|c_1\|^2, \ldots, \|c_{k-1}\|^2)\big),$$

with $\mu_c = (\mu_{c_1}, \ldots, \mu_{c_{k-1}})'$.

In this setting it is hence quite simple to construct (exact) simultaneous tests.

Further testing problems

In the above drug example we had $c_1 = (0, -1, 1)$ and $c_2 = (-1, 1/2, 1/2)$.

These two contrasts are orthogonal. Hence, if we have a balanced design ($n_1 = n_2 = n_3 = m$) we know that under $\mathcal{H}_0$

$$(\hat{\beta}_2 - \hat{\beta}_2, \hat{\beta}_1 - 1/2\hat{\beta}_2 - 1/2\hat{\beta}_3)' \sim N_2(0, \Sigma)$$

where $\Sigma = \frac{\sigma^2}{m}\mathrm{diag}(2, 1.5)$.

Further testing problems

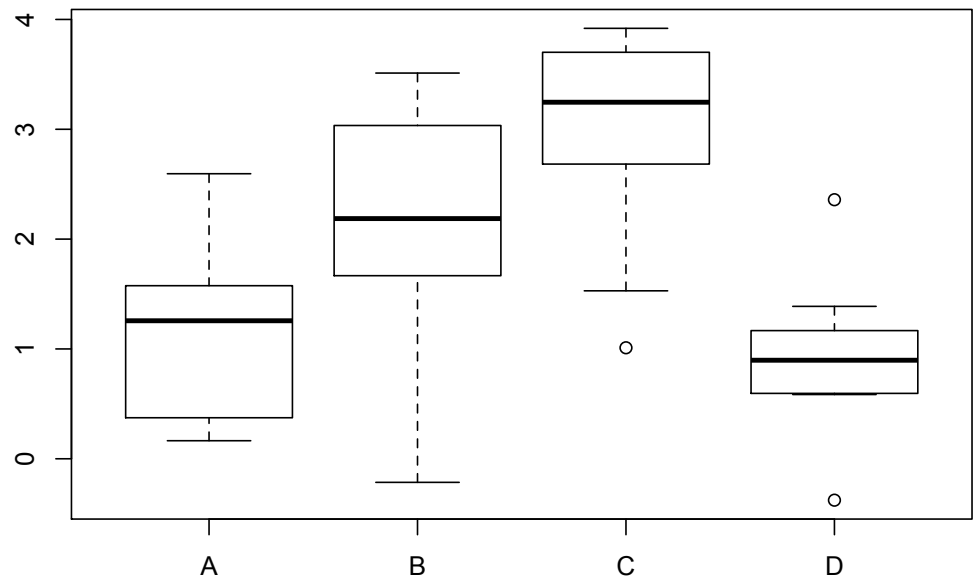
For illustrative purpose we create an artificial example in R with the following layout:

group sizes	n_1	n_2	n_3	n_4
group means	β_1	β_2	β_3	β_4

```
> n1 <- 10
> n2 <- 8
> n3 <- 12
> n4 <- 7
> n <- c(n1,n2,n3,n4)
> fac <- as.factor(rep(c("A","B","C","D"),n))
> beta <- c(1,2,3,1)
> mu <- rep(beta,n)
> y <- mu+rnorm(sum(n))
> boxplot(y~fac)
```

Further testing problems

Here is the corresponding boxplot series:



Note that model assumptions hold by construction.

Further testing problems

Let's apply aov:

```
> mod <- aov(y~fac)
> summary(mod)
            Df Sum Sq Mean Sq F value  Pr(>F)
fac          3  27.81   9.271   10.61 4.9e-05 ***
Residuals   33  28.83   0.874
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Further testing problems

The package `gmodels` contains a function `fit.contrast` that is convenient to use.

```
> install.packages(gmodels)
> library(gmodels)
> mod <- aov(y~fac)
> contr.mat <- matrix(c(1,1,1,0,1,-1,0,-1,0,-1,-1,0),nrow=3)
> fit.contrast(mod, "fac", contr.mat)
                     Estimate Std. Error    t value   Pr(>|t|)
fac c=( 1 0 0 -1 )   0.2174293  0.4606463  0.4720092 0.64002656
fac c=( 1 1 -1 -1 ) -0.6549373  0.6278735 -1.0431040 0.30448177
fac c=( 1 -1 0 0 )  -1.0019631  0.4433875 -2.2597911 0.03055911
```

Remark

- *Since we do simultaneous testing critical values should be adjusted.*
- *To have correct size, we need to decide beforehand which contrasts we would like to test.*

Further testing problems

If the F-test indicates that group means are not identical we can use so-called Post-Hoc-Analysis in order to explore which groups are different.

The naive approach would be to run pair-wise comparisons. This, however, will result in $\binom{k}{2}$ tests and accordingly we inflate our Type 1 error, if we do not appropriately adjust the significance level. This is what Tukey's test is doing. It compares all the pairs with adjusted significance level. Since we compare all pairs, it doesn't matter that we do the analysis after seeing data.

The adjusted significance level is obtained from quantiles of the studentized range statistic.

Further testing problems

With the quantiles of this studentized range statistic one can construct simultaneous confidence intervals for $\beta_i - \beta_j$.

NOTE: if the sample sizes in groups are not equal the confidence intervals become too large/conservative.

Further testing problems

At this point, we don't go into further details and just show how to implement in R.

```
> model.sim <- aov(y~fac)
> TukeyHSD(model.sim)
  Tukey multiple comparisons of means
    95% family-wise confidence level
$fac
          diff        lwr         upr     p adj
B-A  1.0019631 -0.1973768  2.20130301 0.1284802
C-A  1.8743297  0.7917205  2.95693899 0.0002625
D-A -0.2174293 -1.4634535  1.02859485 0.9647017
C-B  0.8723666 -0.2816987  2.02643196 0.1926335
D-B -1.2193924 -2.5279795  0.08919468 0.0750969
D-C -2.0917590 -3.2942676 -0.88925047 0.0002463
```

Further testing problems

There are several different post-hoc tests available. This indicates that they all have the one or the other deficiency.

In practice one should carefully check model assumptions and see which test might be suitable for given data.

What one definitely should not do, is to apply a variety of such tests, until one finds support for the own hypothesis ...

Further testing problems

If we cannot reasonably assume normality, then instead of transforming the data, we can adopt rank-based procedures.

Let R_{ij} be the rank scores of observation j from group i (among all observations). We briefly discuss the Kruskal-Wallis-test (KWT).

Let us assume that $Y_{i1}, \ldots, Y_{in_i} \sim F_i$, where F_i is some continuous distribution. Here the index $1 \leq i \leq k$ refers to the i-th group. We test

$$H_0 \colon F_1(z) = \cdots = F_k(z) \quad \text{v.s.} \quad H_A \colon F_i(z) = F(z - \theta_i)$$

and $\theta_i \neq 0$ for some i.

Then the KWT statistic is (up to a multiplicative constant) the F-statistic computed from the rank scores and has an asymptotic χ^2 distribution. For a small number of observations we can rely on a finite sample distribution.

Further testing problems

In fact, we can use this method very generally for data on an ordinal scale. Assuming data within each group are iid we can use the KWT for testing

$$H_0 : ER_{11} = \cdots = ER_{k1} \quad vs \quad H_A : \neg H_0.$$

(Note that the iid assumption within a group implies $ER_{ij} = ER_{ij'}$.)

Though the ranks are not normal, the test will work at least in the asymptotic framework if the homogeneous variance assumption for the ranks is justified.[2]

Note that the test is invariant under monotone transformations.

[2] For a detailed discussion on the exact framing of this test see Vargha and Delaney: The Kruskal-Wallis Test and Stochastic Homogeneity, Journal of Educational and Behavioral Statistics, 1998, Vol. 23, pp. 170–192.

Further testing problems

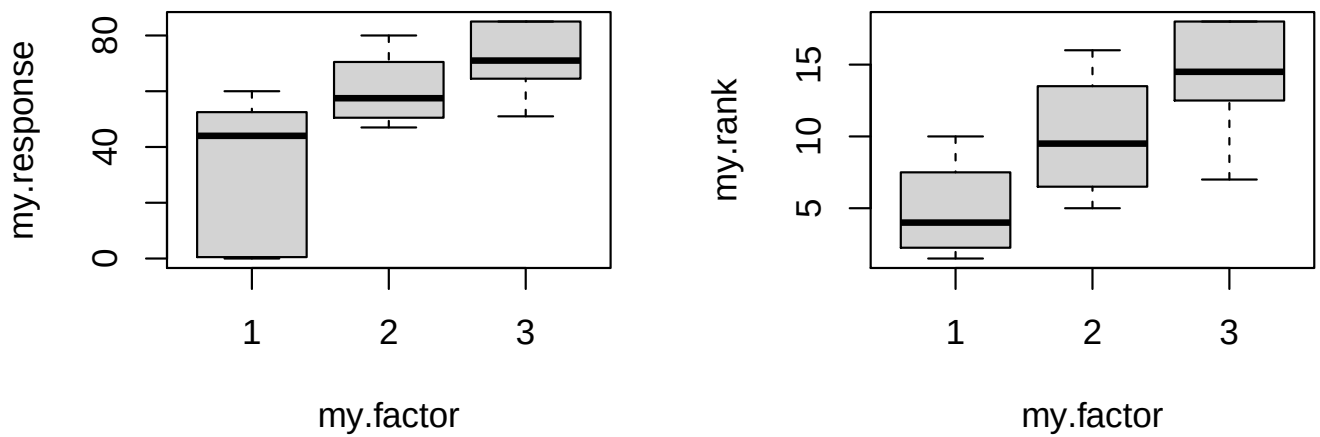
The following data is taken from the internet (`https://www.methodenberatung.uzh.ch/de/`). We want to compare concentration of patients by frequency of taking vitamines.

ID	Gruppe	Konz	Ränge "Kein Vitamin"	Ränge "1xVitamin"	Ränge "4x Vitamin"
2	1	0	1.5		
4	1	0	1.5		
3	1	1	3		
7	1	44	4		
19	2	47		5	
5	1	50	6		
10	3	51			7
17	2	54		8	
1	1	55	9		
6	1	60	10		
16	2	61		11	
11	3	63			12
14	3	66			13
15	3	69			14
8	3	73			15
18	2	80		16	
9	3	85			18
12	3	85			18
13	3	85			18
	Rangsummen:		35	40	115
	Gruppengrösse:		7	4	8

Abbildung 1: Beispieldaten und erste Berechnungsschritte (Gruppe: 1 = Kein Vitamin, 2 = Einmal Vitamin, 3 = Viermal Vitamin)

Further testing problems

```
my.factor=factor(c(rep(1,7),rep(2,4),rep(3,8)))
my.response=c(0,0,1,44,50,55,60,47,54,61,80,51,63,66,69,73,85,85,85)
boxplot(my.response~my.factor)
my.rank=rank(my.response)
boxplot(my.rank~my.factor)
kruskal.test(my.response~my.factor)
Kruskal Wallis chi squared = 10.407 df = 2 p-value = 0.005496
```



The boxplot indicates violation of homogenous variance and normality assumption. The ranks look more homogenous and hence KWT is applicable. KWT indicates a significant difference between groups.